

REMOVING GRAZING PRESSURE FROM A NATIVE PASTURE DECREASES SOIL ORGANIC CARBON IN SOUTHERN NEW SOUTH WALES, AUSTRALIA

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ABSTRACT

Grazing management is a known influence of carbon accumulation in agricultural soil, but there is conflicting evidence on the extent. This study compared organic carbon and nitrogen stocks at the conclusion of a 5-year grazing trial on a fertilised native pasture in south-eastern Australia. The study included three grazing treatments: ungrazed, tactically grazed (set stocking with biannual rest periods) and cell grazed (intense stocking with frequent long rest periods). There was no influence of grazing treatment detected on pasture sward composition when averaged over seasons or on total nitrogen or bulk density. The cell grazing treatment had total carbon stock of 32.9 Mg C ha⁻¹ (SE = 1.8) in the 0–0.30 m soil layer, which was a significant increase ($p < 0.05$) relative to the ungrazed treatment at 25.6 Mg C ha⁻¹ but not statistically greater than the tactical treatment at 29.5 Mg C ha⁻¹. There was no difference detected in labile carbon stocks to 0.30 m, which indicates that differences in soil carbon due to grazing was accumulated over the 5-year trial rather than reflecting short-term seasonal impacts. We propose that a combination of factors contributed to a greater stock of soil carbon under grazed pastures including differences in plant shoot/root allocation, root growth and root turnover with defoliation under grazing as well as lower plant productivity where grazing is excluded because of shading and nutrient tie-up. This study demonstrates removing grazing pressure may lead to lower soil carbon stocks in native pastures over time and provides evidence of the potential for grazing management to increase soil carbon in the short-term. Copyright © 2016 John Wiley & Sons, Ltd.

KEY WORDS: carbon concentration; carbon sequestration; grazing management; perennial pastures; legumes

INTRODUCTION

Increasing the organic carbon (OC) content of soil benefits agricultural production, ecosystem services and the global community through enhanced soil aeration, permeability, available moisture, nutrient cycling and storing carbon (C) that could otherwise be warming the atmosphere (Keesstra *et al.*, 2016; Lal, 2004; Paustian *et al.*, 1997). Pasture management that increases pasture growth and herbage mass for livestock may also increase soil C stocks (Conant, 2010; Conant *et al.*, 2001). Grazing management is therefore a key influence of soil C accumulation. The potential benefits are associated with promoting deep-rooted perennial plants, promoting legume growth, encouraging root turnover (Chen *et al.*, 2015) and litter return to the soil surface, increasing the organic matter (OM)-soil contact through trampling and reducing OM loss by minimising soil erosion through maintaining groundcover (Cerdà & Lavée, 1999; Galdino *et al.*, 2015; Palacio *et al.*, 2014; Sanjari *et al.*, 2008; Sanjari *et al.*, 2009). Controlling animal stocking,

intensity and duration can create a more favourable environment for plant growth, and thereby, OM supply to soil, by enhancing properties such as increased soil hydraulic conductivity and infiltration and by reducing the bulk density (BD) of soil (Sanjari *et al.*, 2008; Sanjari *et al.*, 2009; Sarah & Zonana, 2015). In particular, livestock grazing patterns and trampling routes have a demonstrated impact on the re-distribution of sediments, OM and runoff (Sarah & Zonana, 2015). Grazing management is a practice that land managers can change and may be an important tool to increase soil C under pastures. This study compares soil C stocks under three grazing treatments on a native pasture at the conclusion of a grazing trial as an example of the potential for grazing management to influence C sequestration in soil.

Well-managed perennial pastures are thought to represent the maximum opportunity for agricultural soils to accumulate C (Conant *et al.*, 2003; Ingram *et al.*, 2008; Lal, 2004; Paustian *et al.*, 1997; Sanderman *et al.*, 2010; Sherrod *et al.*, 2005). However, there is little information on what constitutes well-managed perennial pastures in the context of grazing management for soil C accumulation. The relationship between soil C and grazing management is complex, but if grazing management can achieve increased biomass production, increased root turnover and increased

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groundcover and then soil C stocks are likely to increase (Conant, 2010; Eyles *et al.*, 2015; Lorenz & Lal, 2005; Piñeiro *et al.*, 2010; Smith *et al.*, 2014). Primarily, grazing management can have a rapid and substantial influence on pasture composition (Chapman *et al.*, 2003) and therefore change the decomposition of OM supplied to soil through the type, quantity and location of OM deposited in the soil profile (Ganjegunte *et al.*, 2005).

Despite anecdotal evidence of increased soil C stocks under short duration, high intensity forms of grazing management such as cell grazing, there is conflicting evidence regarding the impacts of grazing management on soil C in the literature. There are reports of decreases in soil C with increased grazing intensity (Derner *et al.*, 1997; Frank *et al.*, 1995; Golluscio *et al.*, 2009; Ingram *et al.*, 2008; McSherry & Ritchie, 2013), increases in soil C with increased grazing intensity particularly when light grazing was compared with no grazing (Ganjegunte *et al.*, 2005; Liu *et al.*, 2012; McSherry & Ritchie, 2013) and no difference in soil C regardless of grazing intensity (Kieft, 1994). In particular, seven Australian studies, which compared the impact of grazing management by rotation with continuous grazing on soil C to 0–30 m, detected no difference in soil C stocks (Allen *et al.*, 2013; Chan *et al.*, 2010; Orgill *et al.*, 2014; Pringle *et al.*, 2011; Pringle *et al.*, 2014; Sanderman *et al.*, 2015; Sanjari *et al.*, 2008). Meanwhile, results from the USA showed soil C stocks increased by 8.4 Mg C ha^{-1} (to 0–50 m) after 25 years of rotational grazing compared with an adjacent continuously grazed pasture (Conant *et al.*, 2003). Four explanations of the variability in response of soil C to grazing management can be derived from the literature. First, the definition of the grazing management is important as stocking intensity; duration and stock type are all likely to influence the impact on soil C and soil properties. Second, grazing management may not happen in isolation from other management, such as nutrient management, which may have confounding effects. Third, the response of soil C to grazing management is context-specific and likely to be influenced by climate, soil type and pasture composition. Last, a comprehensive and sophisticated sampling plan is required to detect smaller scale changes that could be attributed to grazing management.

The definition of grazing management is important, and there are challenges when comparing different grazing management categories. For example, both a continuously and rotationally grazed pastures may be under or overgrazed at low or high stocking rates. Continuously grazed fields are commonly referred to as 'set stocked' and grazed without rest for extended periods of time with the exception of removing animals for animal husbandry such as shearing, marking or drenching. The impact of grazing pressure on pastures in a continuously grazed field will depend on factors such as stocking rate, pasture type, distance to watering points, soil type and nutrition, topography, climate, aspect and field size. There are several different types of grazing management, which involve a rotational grazing plan with a period of rest from grazing. Time controlled grazing was

first proposed by Savory in 1978 (Savory & Parsons, 1980) to address desertification problems in Africa and America. The term time controlled grazing is often used interchangeably with the 'Savory Grazing Method', short duration grazing, holistic resource management or cell grazing. While there are subtle differences in these systems, they all rely on moving large numbers of stock around the landscape based on time or pasture growth (Savory, 1983; Savory & Butterfield, 1988). Cell grazing was introduced to Australia in 1989 (McCosker, 2000), and despite adoption across a range of climates and substantial soil C sequestration claims, there is little literature quantifying the impacts on soil C stocks.

There are several problems when comparing grazing management on farms. Farm-based surveys often measure conditions in a continuously grazed field against one of many rotationally grazed fields that are created by the grazing cell (Chan *et al.*, 2010; Conant *et al.*, 2003; Orgill *et al.*, 2014; Sanjari *et al.*, 2008). This can have confounding affects related to the time of year sites are grazed, rested and sampled, as well as grazing pressure and landscape positions in the fields represented. Of particular concern, with C stock calculations, is the influence this may have on soil BD. For example, soil compaction, measured in this case as BD, will be exacerbated when soils are wet and more so on soils with high clay content. Under a continuously grazed treatment stock is present when the soil is at the highest risk of compaction. Under rotational grazing, only a few of the fields are likely to be grazed at this same point in time, and because of rotation schedules and unpredictability of these conditions, these fields may not be sampled.

Grazing management influences pasture composition, pasture growth and herbage mass. The ability of plants to influence soil C accumulation primarily depends on the quantity and nutrient composition of OM supplied to the soil via root turnover, root exudates and organic litter at the soil surface. Cross-sectional studies of herbage mass in systems with completely different temporal patterns of utilisation may deliver misleading conclusions. This also highlights the problems with comparing only one or a subset of grazing cells with a continuously grazed field at a fixed point in time. If that point in time comparison is after a significant period of rest, the herbage growth rates may be similar; however, the point in time sampling may suggest a higher herbage accumulation.

Rotational grazing may be part of a management regime, which includes low or no nutrient inputs from mineral and organic fertilisers. Soil OC is positively correlated with the nitrogen (N), phosphorus (P) and sulfur (S) content of soil (Chan *et al.*, 2010; Orgill *et al.*, 2014). Orgill *et al.* (2014) compared soil C stocks under continuously and rotationally grazed pastures and detected no difference to 0–30 m, and a greater stock of C under continuously grazed pastures when the 0 to 0–70-m soil layer was considered. It was suggested that this may be due to the higher nutrient content in soil under the continuously grazed pastures as a consequence of active soil nutrient programmes compared with low or no nutrient input at the rotationally grazed sites.

The purpose of this study was to compare total C, total N and labile C stocks under three grazing treatments on a fertilised, native pasture at the conclusion of a 5-year grazing trial in the Monaro region. C stocks to 0.40 m were compared under three grazing treatments: ungrazed (grazing exclusion), tactically grazed (rested for 4–6 weeks to allow seed set and recruitment) and cell grazed (heavily grazed and then rested for 80–120 days) on a native pasture where critical P and S targets were addressed with a soil nutrient programme.

MATERIALS AND METHODS

Study Site

This study was conducted within a NSW Department of Primary Industries fertiliser and grazing trial on a commercial sheep property (latitude $-36^{\circ}24'15''$, longitude $148^{\circ}51'30''$, elevation 860 m) near Berridale, south-eastern NSW. The trial was part of the Monaro Research, Development and Demonstration of Sustainable Grassland Management Project 2005–2010 (MRDSGM; SECMB C1/8) (Pope *et al.*, 2011). Berridale has an average annual rainfall of 582 mm (1947–2015), which is summer dominant and average annual maximum and minimum temperatures of 18°C and 4°C (Bureau of Meteorology, 2015). There was below average annual rainfall for the 5 years prior to the trial commencing and for all years of the trial with the exception of 2007 (Figure 1a).

The site was native grassland dominated by native perennial grass species, including wallaby grasses (*Rytidosperma* spp.), speargrasses (*Austrostipa* spp.) and snowgrass (*Poa sieberiana*). Naturalised annual grasses, legumes and weeds were also present. The site had been grazed by sheep since the early-1900s and had never been cultivated. Prior to trial establishment, the site had no fertiliser applied and no introduced pasture species sown. The stocking rate of the field before the experiment was 3 dry sheep equivalent per hectare (DSE ha^{-1}).

The fertiliser and grazing trial ran from 2005 to 2010. The grazing treatments selected for this study were located within a high fertiliser treatment, with subterranean clover (*Trifolium subterraneum*) broadcast before grazing commenced. Initial soil tests reported that OC was $1.37 \text{ g (100 g)}^{-1}$ (Walkley and Black method); available P (Colwell) was 18 mg kg^{-1} , and available S (KCl_{40}) was 2 mg kg^{-1} in the surface 0.10-m soil layer. The high fertiliser treatment aimed to rapidly achieve the Colwell P target ($>30 \text{ mg kg}^{-1}$) (Gourley *et al.*, 2007) with 250 kg ha^{-1} of single superphosphate applied in autumn annually after which the maintenance P was applied based on the stocking rate. In 2006, gypsum (125 kg ha^{-1}) was applied to address S deficiencies.

The soil is a Lixisol (World Reference Base for Soil Resources; IUSS Working Group WRB, 2015; Brown Chromosol, Australian Soil Classification) derived from biotite granodiorite (White *et al.*, 1976). The A1 horizon (0–0.08 m) is a moderately structured, brown to greyish brown sandy loam (10–20% clay), with a pH of 5.0 in 0.01 M

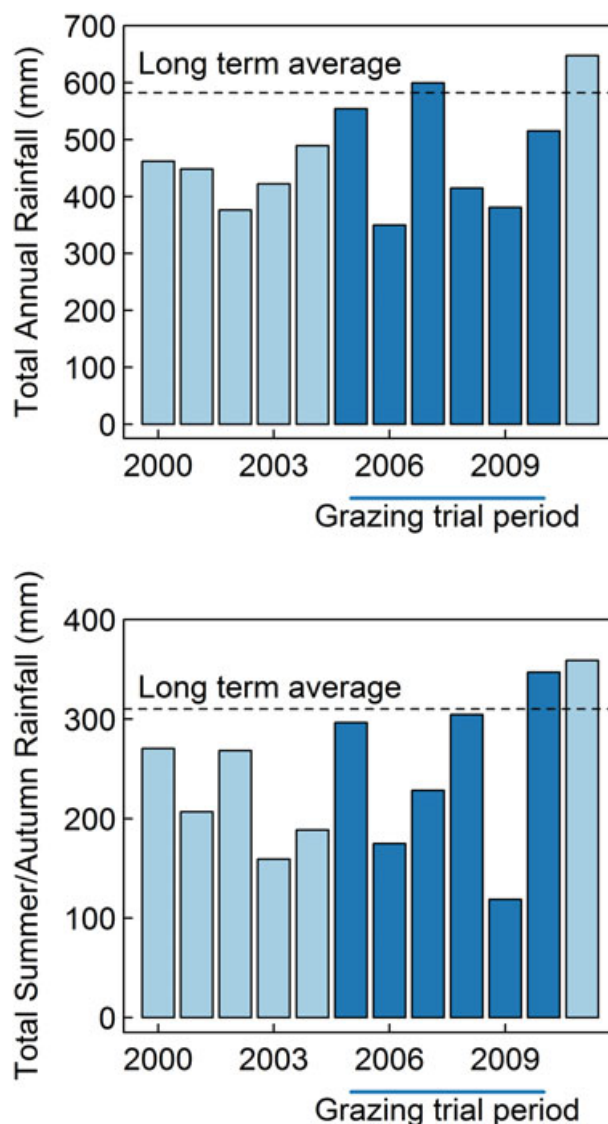


Figure 1. Annual rainfall and summer plus autumn: December–May rainfall (mm) for the Berridale grazing trial site for 2000–2011. Data were taken from the Berridale Bureau of Meteorology station (071022) and incomplete years (2000–2006) substituted by the nearest station, Cooma airport (070217). Long-term average is from the Berridale Bureau of Meteorology station (071022) and calculated from 1947 to 2015. This figure is available in colour online at wileyonlinelibrary.com/journal/ldr.

CaCl_2 and cation exchange capacity (CEC) of $5.8 \text{ cmol} + \text{kg}^{-1}$. The A2 horizon (0.08–0.32 m) is a strongly bleached, poorly structured light sandy clay loam (10–20% clay), with a pH (CaCl_2) of 5.3 and CEC of $3.6 \text{ cmol} + \text{kg}^{-1}$. The B2 horizon (0.32–0.45 m) is a moderately structured, yellow brown and medium clay ($>35\%$ clay), with a pH (CaCl_2) of 5.5 and CEC of $11.1 \text{ cmol} + \text{kg}^{-1}$. The C horizon starts at approximately 0.45 m with a pH (CaCl_2) of 5.5 and CEC of $14.1 \text{ cmol} + \text{kg}^{-1}$.

Grazing

Grazing treatments, ungrazed, tactically grazed and cell grazed, were replicated across three fields that had similar soil-landscape attributes and located within 500 m of one another. The grazing treatment plots (three small fenced

areas in each field; 10 m × 10 m) were established in October 2005 and grazed by Merino wethers. The ungrazed plots were excluded from grazing for the duration of the trial. The tactically grazed plots were rested for 4–6 weeks over late spring/summer and late summer/autumn to allow seed set and recruitment. The cell-grazed plots were grazed four to five times per year by mob stocking, with long and variable rest periods between grazing. Every 80–120 days, the cell-grazed plots were grazed for 1 day at a stocking density that was estimated at approximately 300–400 DSE ha⁻¹ during grazing and an average annual stocking rate that represented the background stocking rate of the field.

Wethers were weighed and fat scored and then allocated to the high fertility fields so that mean commencing weights and fat scores on each field were similar. Stocking rates commenced at district average (3 DSE ha⁻¹) and were increased (up to 5 DSE ha⁻¹) as required with a goal of keeping a consistent fat score across treatments so that increases in productivity were reflected in increased animal product per hectare rather than per head. Additional animals used to increase stocking rates were from the same flock at the same site. Due to periods of drought during the experiment, when weights and fat scores of animals indicated they were being unduly stressed and/or pasture groundcover was being compromised, animals were removed until conditions improved. These destocking events occurred on several occasions (Pope *et al.*, 2011).

Botanical composition (% and kg ha⁻¹) and ground cover (%) were assessed annually in spring in ten 0.5 m × 0.5 m quadrats along fixed transects in each grazing treatment plot using the BOTANAL technique (Tohill *et al.*, 1992). Botanical composition data based on % of dry matter is presented.

Soil Sample Collection and Analytical Methods

Soil chemical properties were measured in 2009 (Table I). Soil samples for C and N analysis were collected in August 2011 after the conclusion of the grazing trial. Soil cores were collected using a hydraulic soil corer to a depth of at least 0.40 m and divided into five soil layers at intervals of 0–0.05, 0.05–0.10, 0.10–0.20, 0.20–0.30 and 0.30–0.40 m.

Two cores from each plot were carefully collected using a 0.075-m diameter core for BD measurements. BD was determined on each soil layer from samples dried at 105°C as described by Dane & Topp (2002). Results were

calculated as BD in Mg m⁻³ on an oven-dry basis to the nearest 0.01 Mg m⁻³.

For soil chemical analysis, five 0.040-m diameter cores were collected per plot. Soil from each soil layer was composited in the field to produce one representative sample per soil layer from each plot. Soil samples were oven-dried at 40°C, passed through a 2-mm sieve and gravel weight recorded and then ground to <0.5 mm using a single puck mill head (Rayment & Higginson, 1992; Method 1B1).

All soil chemical analyses were conducted according to Rayment & Lyons (2011): Colwell Phosphorus (P Col mg kg⁻¹; Method 9B2), Extractable Sulfur (S KCl₄₀ mg kg⁻¹; Method 10D1), Cation Exchange Capacity (CEC cmol_c kg⁻¹; Method 15E1) and soil pH (pH CaCl₂; Method 4B2).

Samples were tested for inorganic C using hydrochloric acid and observing the degree of effervescence (Rayment & Lyons, 2011; Method 19D1). No samples required pre-treatment for inorganic C prior to total C analysis. Total C and total N were determined on all soil samples on approximately 2 g of finely ground soil using a LECO (CNS 2000) combustion furnace (Merry & Spouncer, 1988; Rayment & Higginson, 1992; Method 6B3). Results for this paper report total C and total N as g (100 g)⁻¹ on an oven-dry basis.

Labile C was determined on all samples using the Blair *et al.* (1995) method which involved oxidation with 333 mM KMnO₄. Results are reported as labile C g (100 g) on an oven-dry basis of soil.

Total C, total N and labile C stocks (Mg ha⁻¹) based on an equivalent mass of soil to 0.30 m were calculated using the BD and C or N concentration from each soil layer. This process used Equations 1 and 2 to calculate the stock to a depth standard (DS) for the soil layers (Equation 1) and then to calculate stock based on an equivalent soil mass (ESM) for the 0- to 0.30-m soil layer (Equation 2). There were no BD measurements from the start of the trial, so the ungrazed treatment was used as the reference soil mass. The equations later are for total C, and the same equations were used for total N and labile C.

$$\begin{aligned} \text{DS C stock} & \left(\text{g} (100 \text{ cm})^{-2} \right) \\ &= \text{C concentration} \left(\text{g} (100 \text{ g})^{-1} \right) \times \text{BD} \left(\text{g cm}^{-3} \right) \\ & \quad \times \text{depth (cm)} \times (1 - \text{proportion gravel}), \\ \text{where C stock} & \left(\text{g} (100 \text{ cm})^{-2} \right) = \text{C stock} \left(\text{Mg C ha}^{-1} \right). \end{aligned} \quad (1)$$

Table I. Soil characterisation data (collected in 2009) for the trial site, including total carbon (total C g (100 g)⁻¹), total nitrogen (total N g (100 g)⁻¹), soil pH (pH 1:5 CaCl₂), available P (P Col mg kg⁻¹), extractable S (S KCl₄₀ mg kg⁻¹) and cation exchange capacity (CEC cmol+ kg⁻¹)

Depth (m)	Total C (g (100 g) ⁻¹)	Total N (g (100 g) ⁻¹)	pH (CaCl ₂)	P Col (mg kg ⁻¹)	S (mg kg ⁻¹)	CEC cmol+ kg ⁻¹
0–0.05	1.40	0.10	4.84	58	36	4.9
0.05–0.10	0.92	0.06	5.08	32	31	4.8
0.10–0.20	0.62	0.03	5.28	19	20	4.2
0.20–0.30	0.41	0.01	5.52	–	–	4.8
0.30–0.40	0.25	0.00	5.60	–	–	7.3
0.40–0.50	0.23	0.00	5.43	–	–	12.8
0.50–0.70	0.17	0.00	5.43	–	–	15.5

$$\text{ESM C stock (Mg C ha}^{-1} \text{ 0–0.30 m)} = \text{DS}_{0-20} + \left(\text{DS}_{20-30} \times (\text{ESM}_{\text{ref}} - \text{Soil Mass}_{0-20}) (\text{Soil Mass}_{20-30})^{-1} \right) \quad (2a)$$

Where: $(\text{ESM}_{\text{ref}} - \text{Soil Mass}_{0-20}) (\text{Soil Mass}_{20-30})^{-1} > 1$, the Equation became:

$$\text{ESM C stock (Mg C ha}^{-1} \text{ 0–0.30 m)} = \text{DS}_{0-20} + \left(\text{DS}_{20-30} \times (\text{ESM}_{\text{ref}} - \text{Soil Mass}_{0-20}) (\text{Soil Mass}_{20-30})^{-1} \right) + \left(\text{DS}_{30-40} \times \left(1 - (\text{ESM}_{\text{ref}} - \text{Soil Mass}_{0-20}) (\text{Soil Mass}_{20-30})^{-1} \right) \right) \quad (2b)$$

Where:

DS_{0-20} is the sum of C stock for all soil layers to 0–20 m calculated using Equation 1

DS_{20-30} is the C stock for the 0–20- to 0–30-m soil layer calculated using Equation 1

ESM_{ref} is the reference mass of soil for the 0- to 0–30-m soil layer (ungrazed treatment for each replicate)

Soil Mass_{0-20} is the actual mass of the 0- to 0–20-m soil layer

Soil Mass_{20-30} is the actual mass of the 0–20- to 0–30-m soil layer

DS_{30-40} is the C stock for the 0–30- to 0–40-m soil layer calculated using Equation 1

Statistical Analysis

The pasture composition data were examined by fitting univariate linear models to describe variability in the responses of interest as a function of the experimental design factors. Pasture composition responses were defined by perennial grasses, legumes and annual grasses, and the total contribution to the sward was calculated by the sum of the species components within each group. The most prevalent species within the perennials and legumes were also modelled individually.

Variation in each measure of pasture composition was described by a linear model, which estimated effects due to grazing treatment, season and their interaction. The models included random effects of field replicate and plots within replicates to accommodate potential spatial variation and correlation between repeat measures on the same plot. Null hypothesis significance tests with respect to the main effects and interaction between treatment and season were constructed by *F*-ratio tests. Statistical significance of the effect was concluded when the probability of the observed *F*-ratio under the null hypothesis was less than 5%. The models delivered estimates of mean response to treatment for each season along with standard errors of the means. Pairwise comparison of the means after rejection of the relevant null hypothesis was made using an estimate of least significant difference at 5% critical value. The assumption of normally, identically and independently distributed model residuals was checked by inspection of diagnostic graphs and found to be met for all models.

The soil C, N and BD observations were studied by a series of univariate linear models similar to those used for pasture composition but with depth substituting for seasons. All statistical models and graphical presentations were constructed in the R environment (R Core Team, 2015).

RESULTS

Pasture Composition

When the functional groups of annuals, perennials and legumes were considered, grazing treatments did not have a statistically important effect on the sward composition when averaged over seasons (Figure 2). However, there was a significant ($p < 0.05$) interaction between season and grazing treatment effect on the annual grass composition. This was due to tactically grazed pastures having a greater proportion of annual grasses than the cell and ungrazed treatments in 2006 (Figure 2). This response is likely due to their first closure in autumn 2006 as part of the treatment plan. The null hypothesis for equal contribution over seasons was rejected for perennials and legumes and approached the $p < 0.05$ rejection level for annuals ($p = 0.07$). From 2005 to 2009, there was no grazing treatment effect on pasture composition. There was a statistically significant ($p < 0.05$) effect of season on pasture composition in 2010, coinciding with above average summer and autumn rainfall (Figure 1b) and resulting in increased legumes and consequent decline in perennial grasses (Figure 2). The perennial grass, annual grass and legume species were primarily one of four species: *Rytidosperma* spp. (wallaby grass) and *Austrostipa* spp. (spear grass) for the perennial grass species, *Vulpia* spp. (silver grass) for the annual grass species (although annual grasses represented less than 5% of the total pasture composition) and *Trifolium arvense* (hare's foot clover) for the legume species (Figure 3).

Total Carbon, Total Nitrogen, Labile Carbon and Bulk Density to 0–40 m

Soil depth significantly influenced total C, total N, labile C and BD for all grazing treatments ($p < 0.01$). This reflected

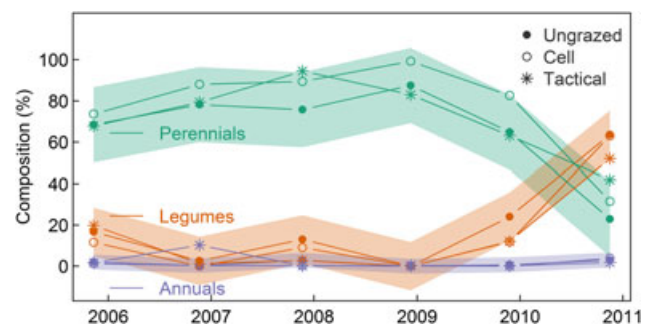


Figure 2. Percentage of perennial grasses (green), annual grasses (purple) and legumes (orange) for the ungrazed, cell grazed and tactically grazed treatments. Samples collected in Spring 2005–2010. An approximate 95% confidence region for the ungrazed treatment is indicated by shading. This figure is available in colour online at wileyonlinelibrary.com/journal/ldr.

an observed decrease in total C, total N and labile C, and an increase in BD with increasing soil depth (Figure 4). Despite the cell and tactically grazed treatments having observed higher mean total C concentration throughout the profile compared with the ungrazed treatment, grazing treatment was not statistically important in the linear model ($p > 0.05$). There was also no grazing effect detected on total N, labile C or BD (Figure 4).

Carbon and Nitrogen Stocks

The hypothesis of equal average total C stock in the 0- to 0.30-m layer across all grazing treatments approached the traditional 5% rejection level ($p = 0.08$). This reflected an observed increase in total C stock in the 0- to 0.30-m soil layer (calculated on an ESM basis) under the cell grazed treatment compared with the ungrazed treatment: 32.9 vs $25.6 \text{ Mg C ha}^{-1}$, respectively (Table II). However, the estimated difference in total C stocks between the cell and

tactically grazed treatments, 32.9 vs $29.5 \text{ Mg C ha}^{-1}$, respectively, was not statistically important. When total C stocks for individual soil layers were compared (using Equation 1: DS calculation), the effect of grazing treatment was statistically important ($p < 0.05$) and reasonably consistent across all soil layers (Table II).

There was no effect of grazing treatment detected in total N or labile C stock (Figure 4) with grazing treatment in individual soil layers to 0.40 m (DS calculation) or in the 0- to 0.30-m soil layer based on an ESM (Table II).

DISCUSSION

The documented effects of grazing management on soil C are often confounded with other site variables, such as soil fertility, that influence OM supply, loss and decomposition. In order to isolate the influence of grazing management on soil C, this trial was located on a site where both critical soil P and S targets were addressed and maintained for the pasture species and production goals. Comparing soil C stocks at the conclusion of the grazing trial indicated that cell grazing was associated with higher soil C stocks in the 0- to 0.30-m soil layer calculated on an equivalent mass of soil compared with the ungrazed treatment; however, there was no difference detected between cell and tactically grazed treatments (Table II). As there were no initial soil measurements, this assumes that the plots had similar initial soil C stocks. Assuming the treatments diverged from the same unknown average initial C stock, implementing grazing as opposed to grazing exclusion resulted in a difference in soil C of $7.30 \text{ Mg C ha}^{-1}$ or $1.46 \text{ Mg C ha}^{-1} \text{ year}^{-1}$ under cell grazing, and $3.90 \text{ Mg C ha}^{-1}$ or $0.78 \text{ Mg C ha}^{-1} \text{ year}^{-1}$ under tactical grazing. Although the difference in soil C stocks under the ungrazed and tactically grazed treatments was not statistically significant. Groundcover at the site was maintained above 80% in all plots for the duration of the trial so the effects of soil erosion, and C redistribution were assumed to have been minimal.

The results from this study are contrary to other studies in southern Australia, which used field surveys to compare the influence of continuous and rotational grazing management on soil C stocks and reported no difference (Chan *et al.*, 2010; Orgill *et al.*, 2014). The estimated sequestration rate from this study for cell grazing is also considerably higher than other estimates of $0.35\text{--}0.41 \text{ Mg C ha}^{-1} \text{ year}^{-1}$ with grazing management (Conant *et al.*, 2003; Conant *et al.*, 2001). There was no significant difference in labile C stocks between the grazed and ungrazed treatments that would support the notion of a sudden increase in recent OM addition to soil (Table II); therefore, the above average summer and autumn rainfall in the final year of the trial (Figure 1b) was not thought to be an important factor contributing to the differences detected.

Soil C is generally not distributed evenly throughout the soil profile. There is usually a greater concentration and stock of C in the surface soil compared with deeper soil

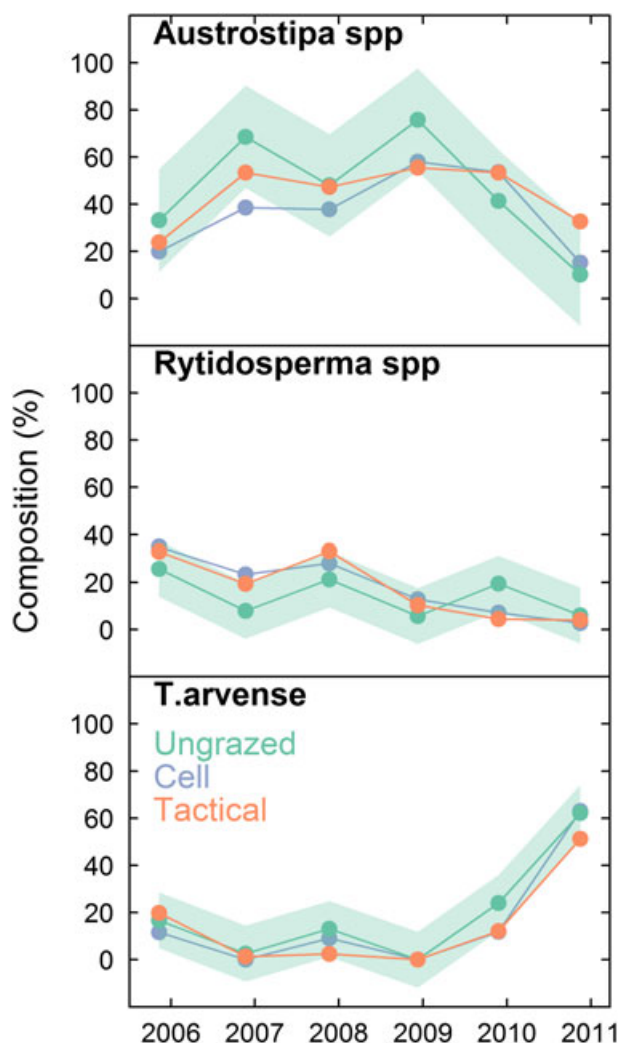


Figure 3. Percent composition for *Austrostipa* spp., *Rytidosperma* spp. and *Trifolium arvense* for the ungrazed (green), cell grazed (purple) and tactically grazed (orange) treatments observed each Spring 2005–2010. An approximate 95% confidence region for the ungrazed treatment is indicated by shading. This figure is available in colour online at wileyonlinelibrary.com/journal/ldr.

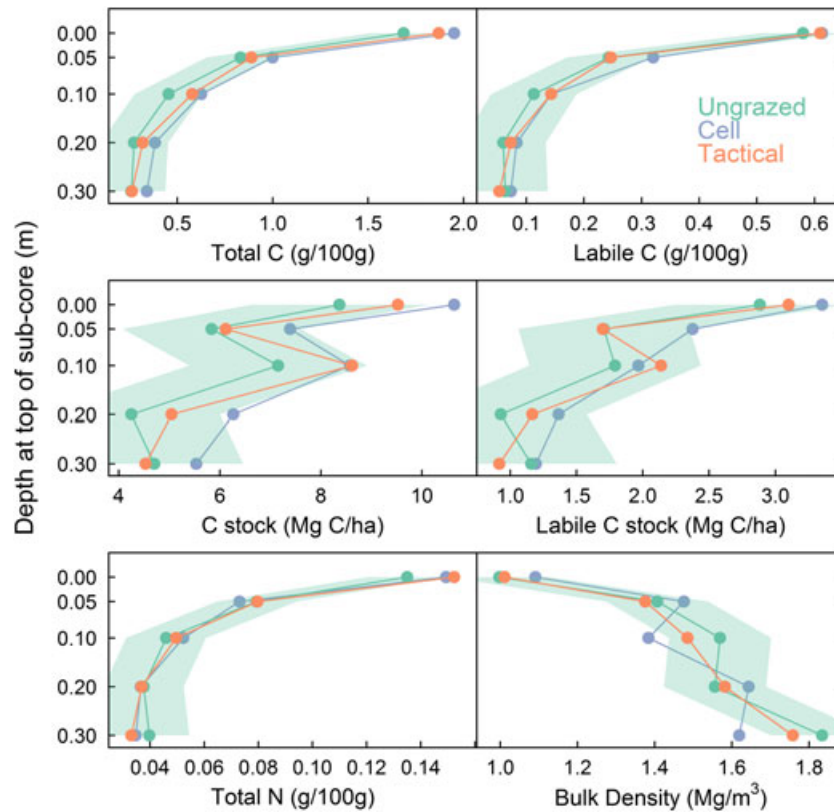


Figure 4. Mean total C concentration (g (100 g)^{-1}), labile C concentration (g (100 g)^{-1}), total C stock (Mg C ha^{-1}), labile C stock (Mg C ha^{-1}), total N concentration (g (100 g)^{-1}) and bulk density (Mg m^{-3}) to 0–40 m for the ungrazed (green), cell grazed (purple) and tactically grazed (orange) plots in Spring 2011. An approximate 95% confidence region for the ungrazed treatment is indicated by shading. This figure is available in colour online at wileyonlinelibrary.com/journal/ldr.

layers. The surface soil layers are typically more sensitive to land management than the deeper soil layers. In this study, the cell grazed treatment had a greater stock of C than the ungrazed treatment (calculated to a DS) in all soil layers, except the 0.30- to 0.40-m soil layer, and there was no

difference from the tactically grazed treatment detected. Increases in soil C with cell grazing are suggested to result primarily from livestock trampling and manure (Southorn, 2002), meaning the increase would be detected in the surface soil layer. This study observed a greater stock of C in

Table II. Mean total C, total N and labile C stock (Mg ha^{-1}) for the 0- to 0.30-m layer (in 2011) calculated on an equivalent soil mass (ESM) and in soil layers to 0.40 m calculated to a depth standard. Least significant difference (LSD), standard error of means (SEM) and *F*-ratio test probability for treatment presented

Carbon and nitrogen stocks (Mg ha^{-1})	Soil depth (m)	Grazing treatment			LSD	SEM	Treatment <i>F</i> -ratio test
		Ungrazed	Tactical	Cell			
Total C stock	0–0.30 (ESM)	25.6	29.5	32.9	7.1	1.8	0.08
Total N stock	0–0.30 (ESM)	2.5	2.7	2.7	0.7	0.2	0.84
Labile C stock	0–0.30 (ESM)	7.3	8.1	9.1	2.9	0.7	0.32
Total C stock – soil layers	0–0.05	8.4	9.5	10.6	3.4	0.9	0.02
	0.05–0.10	5.8	6.1	7.4			
	0.10–0.20	7.2	8.6	8.6			
	0.20–0.30	4.2	5	6.3			
	0.30–0.40	4.7	4.5	5.5			
Total N stock – soil layers	0–0.05	0.7	0.8	0.8	0.3	0.1	0.98
	0.05–0.10	0.6	0.5	0.5			
	0.10–0.20	0.7	0.7	0.7			
	0.20–0.30	0.6	0.6	0.6			
	0.30–0.40	0.7	0.6	0.6			
Labile C stock – soil layers	0–0.05	2.9	3.1	3.4	1.3	0.3	0.30
	0.05–0.10	1.7	1.7	2.4			
	0.10–0.20	1.8	2.1	2			
	0.20–0.30	0.9	1.2	1.4			
	0.30–0.40	1.2	0.9	1.2			

all soil layers to 0.30 m under cell grazing, which coincided with the soil layer immediately above the B horizon (approximately 0.30–0.45 m). Thus, the 0- to 0.30-m zone may be accumulating dissolved OC and other nutrients leaching through the soil profile and may be a zone of lateral water flow or lateral root growth particularly when the B horizon is at wilting point. However, the trial design had grazing plots adjacent to one another so these inherent soil property effects are assumed to be similar for all grazing treatments. When the influence of grazed (cell and tactically grazed) and ungrazed pastures are compared, these results are consistent with Liu *et al.* (2012) who demonstrated that SOC was lower where grazing is excluded. It is likely that a combination of factors are contributing to a greater stock of C in soil under grazed pastures, including differences in plant shoot: root allocation, root growth and root turnover with defoliation under grazing (Fulkerson & Donaghy, 2001), as well as lower plant productivity where grazing is excluded because of shading and nutrient tie-up.

Given the link between grazing management, biomass production and OM supply to soil, if grazing management increases the supply of OM in soil, it is likely to also increase the labile fraction of C in soil. Labile C is partially decomposed OM with a faster turnover time than more stable forms of OM in soil, such as humus and char. Therefore, it is reported to be a sensitive measure of the impacts of management on soil C (Blair *et al.*, 1995). Interestingly, there was no difference detected in labile C stocks to 0.30 m (calculated on an equivalent mass of soil) or for any soil layer to 0.40 m (DS calculation) (Table II). This indicates the increases in soil C have occurred over a period of years during the experiment rather than short-term seasonal impacts and suggests an additive grazing treatment effect from the 5 years of the experiment. An effect of grazing treatment on annual grass composition was detected in 2006, but no effect of treatment was detected in perennial grass or legume composition over the trial period (Figure 2). This agrees with Dowling *et al.* (2005) who reported no consistent effect of cell grazing on species composition compared with continuous grazing at a range of monitored sites in eastern Australia. There was a significant influence of season on pasture composition with an increase in legume composition, primarily *Trifolium arvense*, a common naturalised legume, in 2010 (Figure 2), corresponding with above average summer and autumn rainfall (Figure 1b). This is a native grassland site, with the majority of the pasture species for 2005–2009 being C3 perennial species, primarily *Austrostipa* spp. and *Rytidosperma* spp. The above average annual rainfall in 2007 and above average summer and autumn rainfall in 2010 followed on from more than 10 years of below average rainfall and drought conditions. It has been suggested that drought conditions may make pasture and soil responses more sensitive to grazing treatment effects because of an increase in pasture utilisation with lower growth and similar livestock numbers between grazing treatments (Earl & Jones, 1996). It must be noted that this trial did not compare a continuously grazed treatment nor were the pastures overgrazed.

The MRDSGM Project reported no significant herbage mass benefits between cell and tactically grazed treatments (Pope *et al.*, 2011). Although they did report an increase in litter cover in the ungrazed treatment compared with the grazed treatments (Pope *et al.*, 2011). Litter contributes to OM in soil and regulates soil temperature and moisture and, thus, OM decomposition. Dense litter and standing dry matter can also shade out emerging seedlings, such as annual grasses and legumes. As soil P and S nutrition were adequate at this site, the benefits of fertilising in terms of biomass production may have been lessened where legume and annual grass growth was impeded by shading, for example, in the ungrazed plots. Seedling emergence may also be influenced by the timing of grazing for the cell and tactically grazed treatments. In this environment, N is not applied in mineral fertiliser form; therefore, the legume component of pastures is important as it provides biologically fixed N for grass growth, as well as being an important source of OM and N for soil C stability (Kirkby *et al.*, 2011). While there was no difference detected in total N stocks at the end of the trial (Table II), N supply and cycling throughout the trial may have been less in the ungrazed plots (Chaneton *et al.*, 1996; Lu *et al.*, 2015; Peoples & Baldock, 2001).

This study did not include a continuously grazed treatment. While continuous grazing is not uncommon in this region, tactical grazing is recommended to encourage seed set and enhance pasture composition. Overgrazing can occur in continuously grazed pastures, poorly timed short duration, high intensity grazing or with a rotational grazing plan. When overgrazing reduces net primary productivity, pasture regeneration or detrimentally changes pasture composition, and then soil C stocks may decline (Chapman & Lemaire, 1993). Despite this grazing trial receiving below average annual rainfall in the majority of years, this study provided evidence that when soil nutrient requirements are addressed, grazing may increase soil C stocks compared with excluding grazing. A contributing factor to these positive results may have been that the trial was on native pastures that are inherently resilient to dry conditions. Introduced pastures under the same climatic and grazing treatments may not have responded in the same way. Thus, research is required to determine the influence of grazing management on soil C under introduced pastures. Finally, the soil at the Berridale trial site is a Chromosol, but essentially can be described as shallow granite-derived soil. Prime grazing land in the Monaro region is located on basalt and deep granite-derived soil, which have inherently higher soil fertility, water holding capacity and concentrations of soil C (Orgill *et al.*, 2014). These soils may have a greater capacity to sequester C in soil with grazing management, or these soils may be closer to soil C saturation and therefore less responsive to grazing management compared with the shallow-granite derived soil included in this study.

CONCLUSION

This study demonstrated that removing grazing is unlikely to increase soil C stocks under native pastures in southern

Australia and that active grazing of pastures can significantly increase soil C stocks. While there was no significant difference in soil C between cell or tactical grazing treatments, cell grazing native pastures significantly increased soil C in the 0- to 0.30-m soil layer compared with the ungrazed treatment on a shallow granite-derived soil. Unlike other grazing studies where grazing management is suggested to increase C stocks in the surface soil as a response to trampling and manure (Southorn, 2002), statistically important increases were only detected when the 0- to 0.30-m soil layer was compared. There was no difference detected in labile C or N stocks due to grazing treatment, which may indicate that the differences in soil C stocks were not from short-term accumulation of particulate OM, such as seasonal pasture growth, but may be from longer-term (5 years) management. Despite lower than average rainfall during the trial, soil C under native pastures on shallow granite-derived soil responded to grazing management. It is unknown whether cell grazing would affect soil C stocks under introduced pastures under similar conditions or in the more fertile and productive basalt-derived and deep granite-derived soils of the Monaro region, which already have higher C stocks. Furthermore, these results are from the conclusion of a replicated grazing trial conducted with small plot treatments, which brings the benefits of controlling environmental factors that influence soil C, such as climate, topography, aspect, soil type, controlling grazing pressure and reducing the spatial variability of soil C. However, grazing management occurs at the farm scale, and particularly for the cell grazing treatment, measuring changes in soil C is likely to be more complex with field size, number of cells, distance to water, and variations in soil properties and pasture composition likely to influence the grazing treatment effects and soil C interactions. This study provides evidence that there is potential for agricultural soils to have higher soil C stocks as a result of grazing management compared with no grazing in the short-term.

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